Subgroup Performance

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

THE CONNECTED WEEK

One readable subgroup-evidence record.

Compute separate rates,weighted totals and missing-evidence limits.

Keep one habit: W1–W3/W5 → selected X/Ytask → project slice record → changed-case return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Improve Generalization. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Learning Curves. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Metrics Match Consequences. Give an example and a mistake that the idea helps you avoid.

Improve Generalization

> **My remembered idea, example and correction**

Learning Curves

> **My remembered idea, example and correction**

Metrics Match Consequences

> **My remembered idea, example and correction**

W1 · Explain the mechanism

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A2/20 errors=10%, B3/5=60%, pooled 5/25=20%, gap 50 percentage points. One changed originalBerror yields 2/5=40%. PictureSun/Shadearithmeticmatches; adding 5 error-freeShadecases yields 3/10 and5/30 pooled, not a model repair. Complete synthetic 25-case audit uses lightingbright/dim, typeAlpha/Beta, cameraA/B/C; spottedpositive. Error countsbright 0/10, dim 5/15, Alpha2/10, Beta3/15, A2/20, B3/5, Cunobserved.

  Recallbright5/5, dim 6/8, Alpha5/5, Beta6/8, A10/10, B1/3, Cunavailable; pooled 11/13. Do not sum overlapping field totals.

- Reference for this exercise: open this week's student lesson, "Subgroup Performance", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Calculate the worked group and pooled rates.

My reasoning and evidence

MAIN LESSON · SECTION 13

W2 · Trace the current experiment

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A2/20 errors=10%, B3/5=60%, pooled 5/25=20%, gap 50 percentage points. One changed originalBerror yields 2/5=40%. PictureSun/Shadearithmeticmatches; adding 5 error-freeShadecases yields 3/10 and5/30 pooled, not a model repair. Complete synthetic 25-case audit uses lightingbright/dim, typeAlpha/Beta, cameraA/B/C; spottedpositive. Error countsbright 0/10, dim 5/15, Alpha2/10, Beta3/15, A2/20, B3/5, Cunobserved.

  Recallbright5/5, dim 6/8, Alpha5/5, Beta6/8, A10/10, B1/3, Cunavailable; pooled 11/13. Do not sum overlapping field totals.

- How does correcting one B error affect its rate?

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Repair the reasoning

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A2/20 errors=10%, B3/5=60%, pooled 5/25=20%, gap 50 percentage points. One changed originalBerror yields 2/5=40%. PictureSun/Shadearithmeticmatches; adding 5 error-freeShadecases yields 3/10 and5/30 pooled, not a model repair. Complete synthetic 25-case audit uses lightingbright/dim, typeAlpha/Beta, cameraA/B/C; spottedpositive. Error countsbright 0/10, dim 5/15, Alpha2/10, Beta3/15, A2/20, B3/5, Cunobserved.

  Recallbright5/5, dim 6/8, Alpha5/5, Beta6/8, A10/10, B1/3, Cunavailable; pooled 11/13. Do not sum overlapping field totals.

- Group C has no evaluated cases. What is its measured error rate?

My reasoning and evidence

MAIN LESSON · SECTION 16

W5 · Plan the next check

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Main A2/20 errors=10%, B3/5=60%, pooled 5/25=20%, gap 50 percentage points. One changed originalBerror yields 2/5=40%. PictureSun/Shadearithmeticmatches; adding 5 error-freeShadecases yields 3/10 and5/30 pooled, not a model repair. Complete synthetic 25-case audit uses lightingbright/dim, typeAlpha/Beta, cameraA/B/C; spottedpositive. Error countsbright 0/10, dim 5/15, Alpha2/10, Beta3/15, A2/20, B3/5, Cunobserved.

  Recallbright5/5, dim 6/8, Alpha5/5, Beta6/8, A10/10, B1/3, Cunavailable; pooled 11/13. Do not sum overlapping field totals.

- Why does equal accuracy not settle whether use is fair?

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected experiment task

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Treat the selected X/Ygroups as the supplied disjoint 14-case comparison with the same error definition. Do not use the main 25-case camera table or invent a causal group difference. Report each count, pooled denominator and the four-case Ysample limitation. No per-class labels are given, so recall is not computable from these aggregate error counts alone.

- Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Apply independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 9

Project · Keep the experiment evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Add **subgroup** tables and a minimum acceptable threshold to the Plant Classifier release criteria.

- The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

MAIN LESSON · SECTION 10

Finish · Judge your evidence

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- - I can explain the mechanism.

- I can show evidence.
- I can handle a changed case.
- I can name a limit.

  Mark each as not yet, with help or independently, and attach supporting work.

- Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Use W2: Check sensitivity for **Subgroup** evaluation. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Use the worked case for **Subgroup** evaluation. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier learning target

REFERENCE · SECTION 5

Reference · Earlier learning target

> Keep this task’s complete inputs, stated rule and support record together.

Task and evidence record

- Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

▼ **Extra practice · Earlier complete case**

REFERENCE · SECTION 6

Extra practice · Earlier complete case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- A has18/20 correct; B has6/10.
- Compute **subgroup** results and **evidence** limits.
- Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ Extra practice · Changed case

REFERENCE · SECTION 7

Extra practice · Changed case

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Accuracy is 18/20 in bright light and 6/10 in dim light. Compute both rates and the percentage-point gap.
- My independent answer and reason

> **Write or draw your reasoning and evidence.**

Week 31 Test and reflect

▼ Extra practice · Design a revealing test

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Extra practice · Earlier transfer**

REFERENCE · SECTION 11

Extra practice · Earlier transfer

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Device A gets 18/20 correct; B gets 6/10. Compute overall accuracy and the **subgroup** gap. What causal claim is not supported?

- Record support used: none, hint or teacher explanation.

My independently reasoned answer

▼ **Reference · Earlier W4 record**

REFERENCE · SECTION 15

Reference · Earlier W4 record

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Groups X 1 error/10 and Y 2/4: calculate separate and pooled rates, then state one limit.

My reasoning and evidence

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 18

Extra practice · Pictured case

> Keep this task's complete inputs, stated rule and support record together.

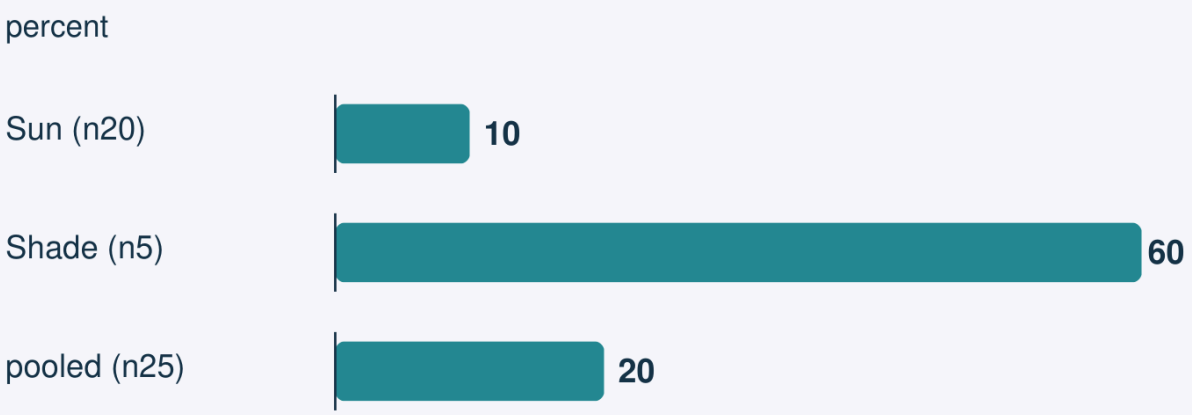
Task and evidence record

- Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.
- Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.
- All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.
- Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

AI ACADEMY / AI-4/31

Subgroup error rates



Bar length shows the supplied value; the marked baseline is zero.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-4/31

Subgroup Performance

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Sun rate

Sun error=2/20=10%.



2. Shade rate

Shade error=3/5=60%.



3. Pooled rate

Total error=5/25=20%, which hides the 50-percentage-point subgroup difference.

CHANGE ONE THING

Shade gains5 new cases with no errors.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 19

Reference · Project portfolio milestone

> Keep this task's complete inputs, stated rule and support record together.

Task and evidence record

- Project: Responsible Plant Classifier
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

  Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict, check and repair

REFERENCE · SECTION 20

Extra practice · Predict, check and repair

> Keep supported feedback separate from the reserved independent case.

First supported question and hint

- Calculate **subgroup** and pooled error rates without letting the larger group hide a gap.
- Which labelled step matches this detail? Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.
- 1. Pooled rate
- 2. Sun rate
- 3. Shade rate
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the first explanation**

After your attempt, compare with the supported explanation: Pooled rate Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Read the picture in order.

Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Changed-case question and hint

- Changed-case practice: Shade gains5 new cases with no errors. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check the changed explanation**

After your attempt, compare with the supported explanation: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Repair and keep the record

Review Improve Generalization (ai-4/30). Goal: Plan one **evidence**-led change and test it on source-separated validation cases. Short practice: What fresh **evidence** is needed after adding crossed **training** data? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Learning Curves (ai-4/28). Goal: Read **training**-size learning curves. Short practice: What comparison is valid if the validation set changes between sizes?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Metrics Match Consequences (ai-4/25). Goal: Apply a declared score threshold including equality. Short practice: Which labels determine the false-alarm count at threshold 0.4?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

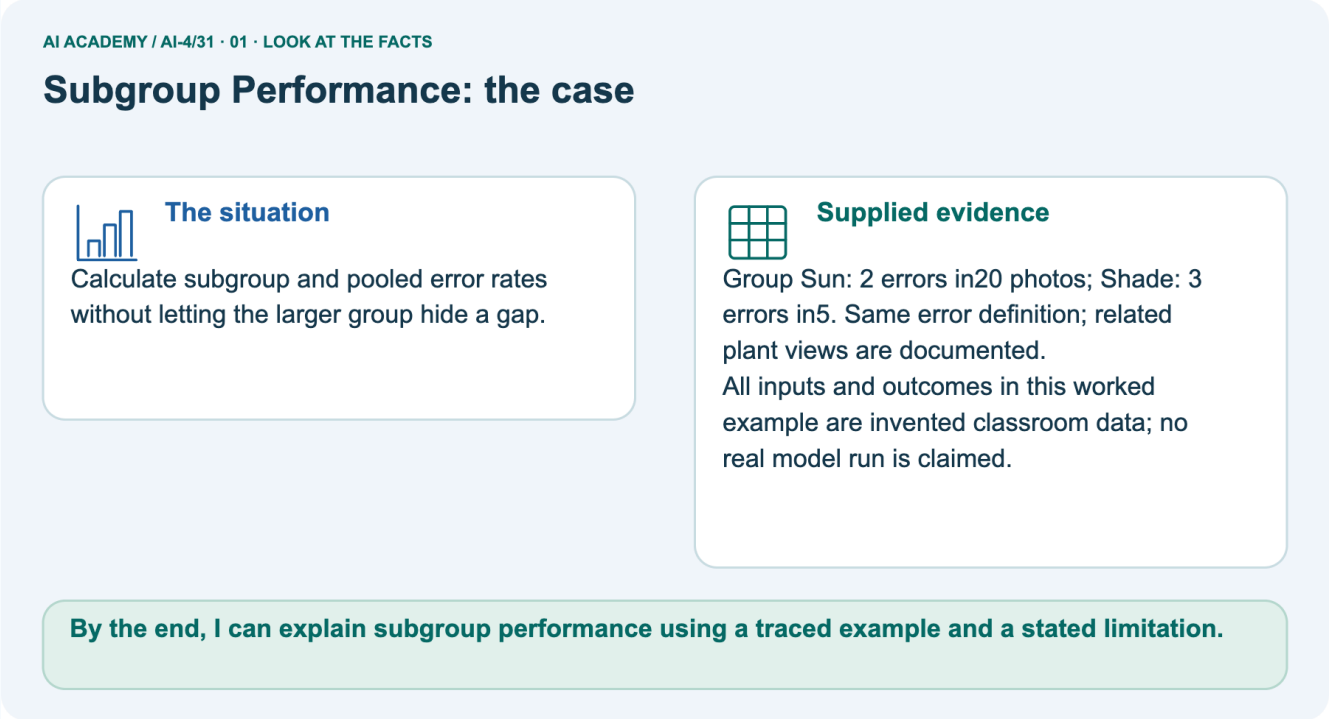
▼ Reference · Six-picture response sheets

REFERENCE · SECTION 21

Reference · Six-picture response sheets

Choose the relevant picture instead of repeating every task.

01. Subgroup Performance: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

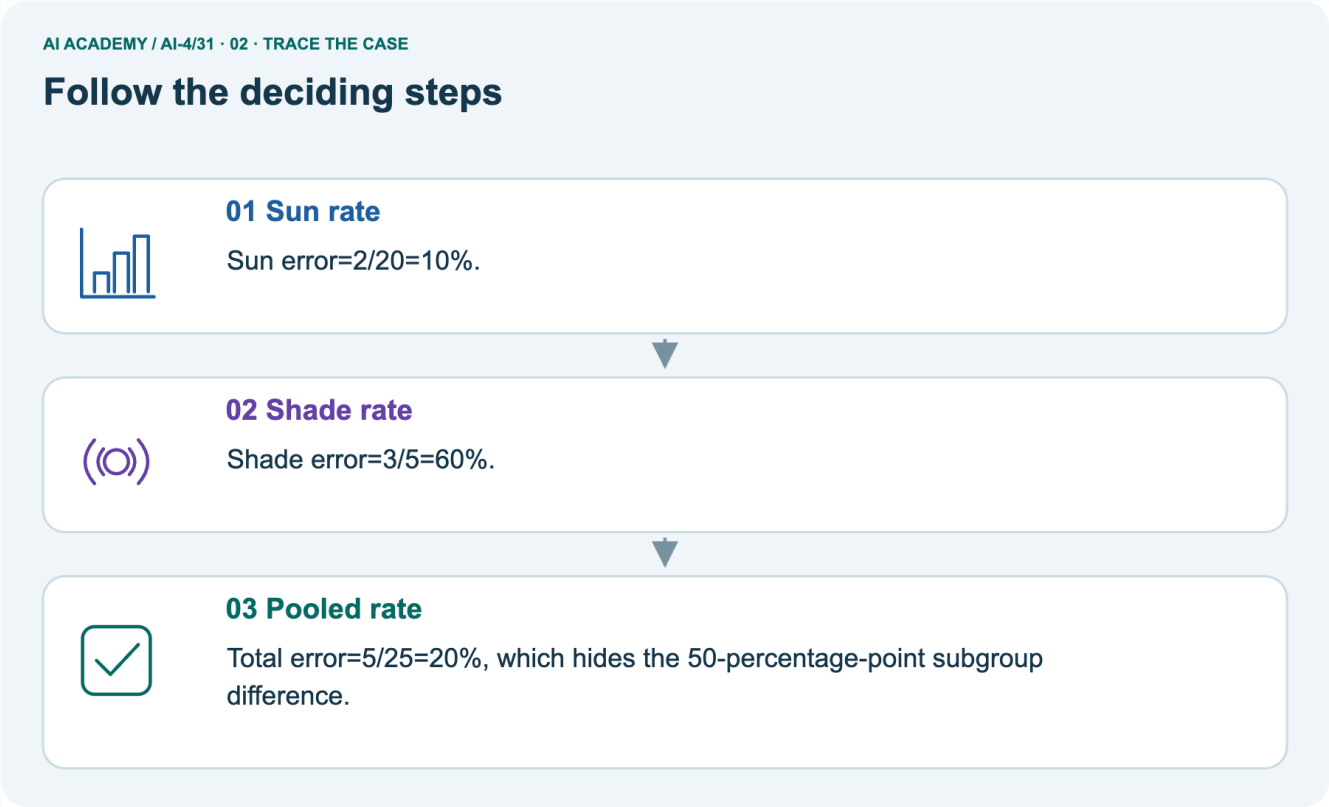
Calculate **subgroup** and pooled error rates without letting the larger group hide a gap. Supplied facts: Group Sun: 2 errors in20 photos; Shade: 3 errors in5. Same error definition; related plant views are documented.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Sun rate: Sun error=2/20=10%. → Shade rate: Shade error=3/5=**60%**. → Pooled rate: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Subgroup error rates



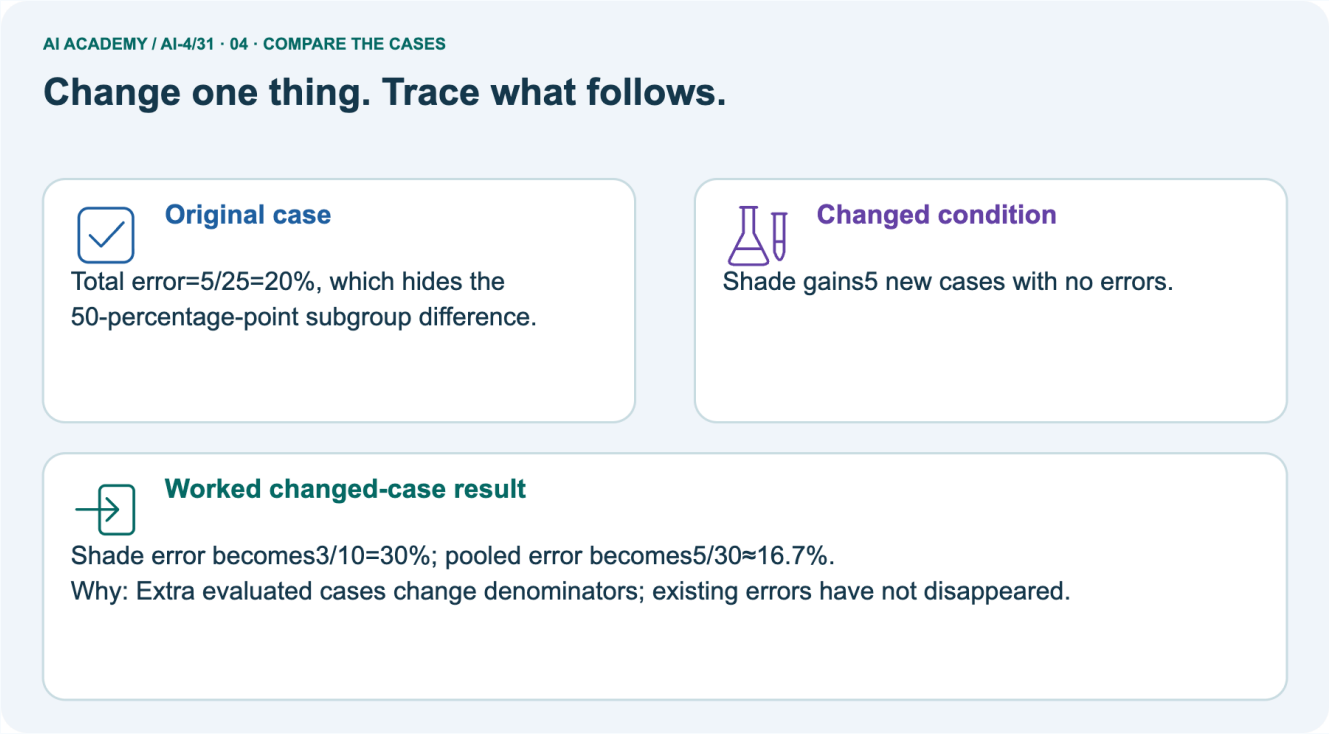
Follow the labelled picture with the matching explanation. Open the vector diagram

Subgroup error rates. Units: percent. Sun (n20): 10. Shade (n5): 60. pooled (n25): 20.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

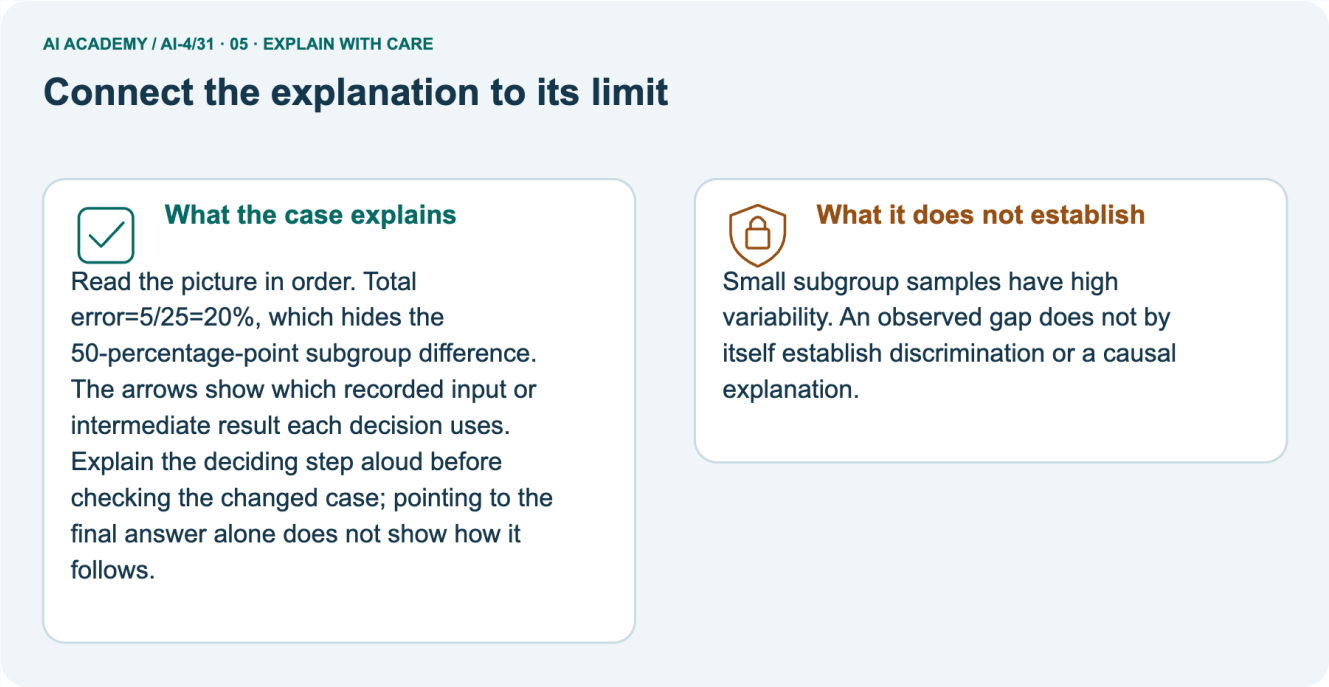
Original case: Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. Changed condition: Shade gains5 new cases with no errors.

Expected result: Shade error becomes3/10=30%; pooled error becomes5/30≈16.7%. Reason: Extra evaluated cases change **denominator**s; existing errors have not disappeared.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Total error=5/25=20%, which hides the 50-percentage-point **subgroup** difference. The arrows show which recorded **input** or intermediate result each decision uses.

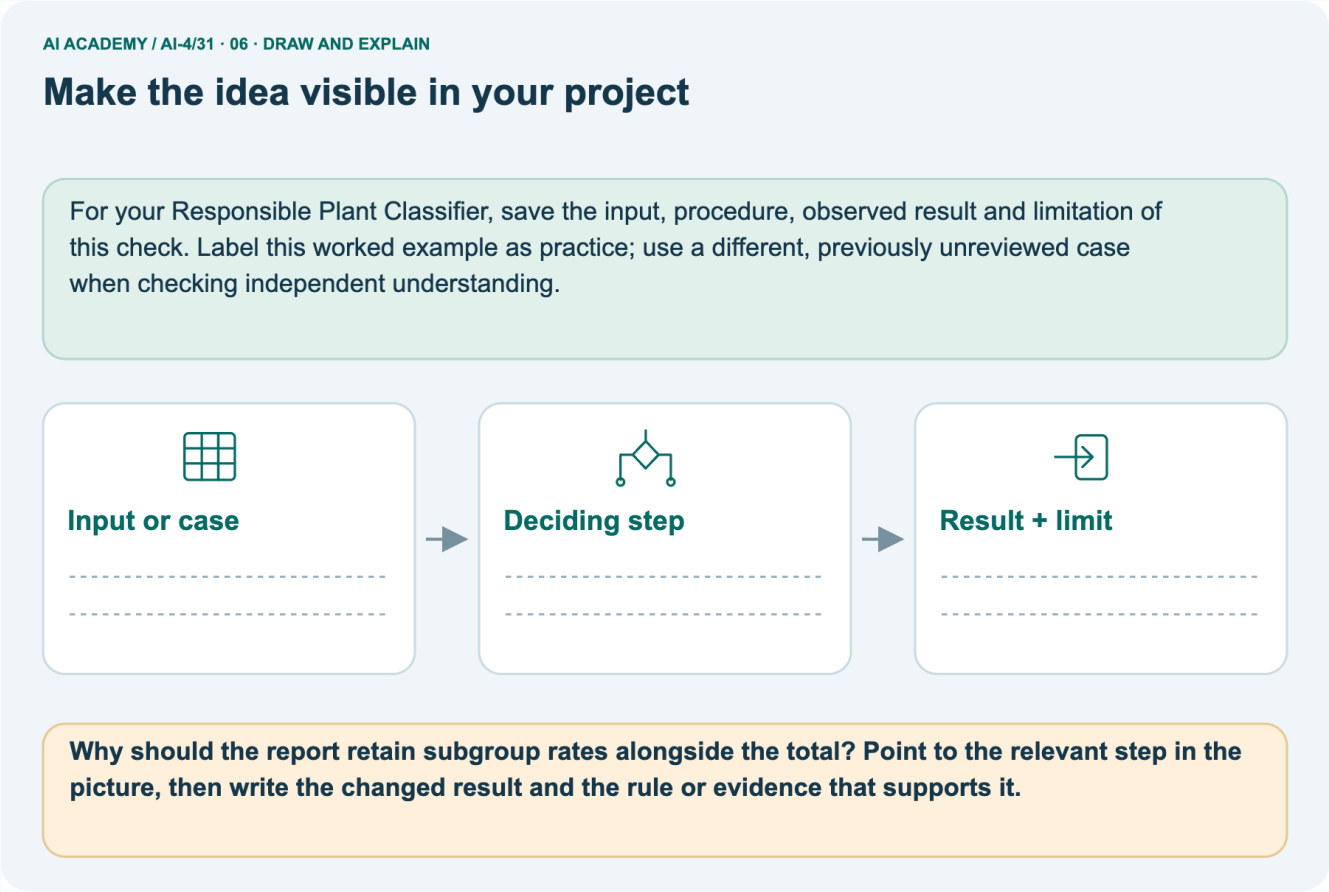
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows. Limit: Small **subgroup** samples have high variability.

An observed gap does not by itself establish discrimination or a causal explanation.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: Why should the report retain **subgroup** rates alongside the total? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Subgroup Performance: the case

02. Follow the deciding steps

03. Subgroup error rates

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.